NACS 645 – Two systems to decide

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Valentin Guigon

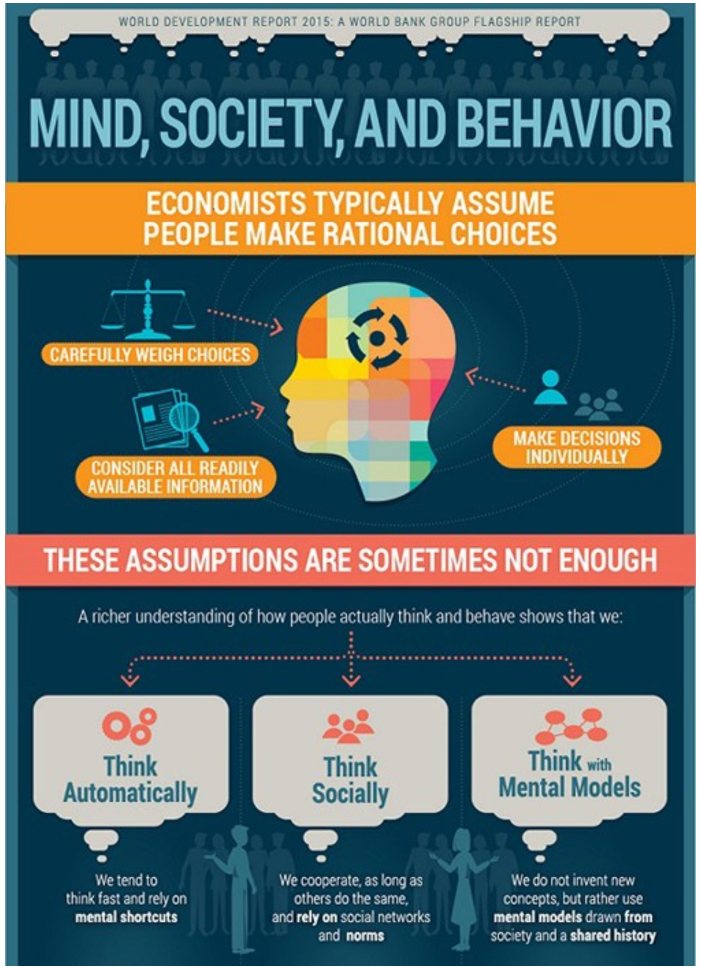
DEPARTMENT OF PSYCHOLOGY

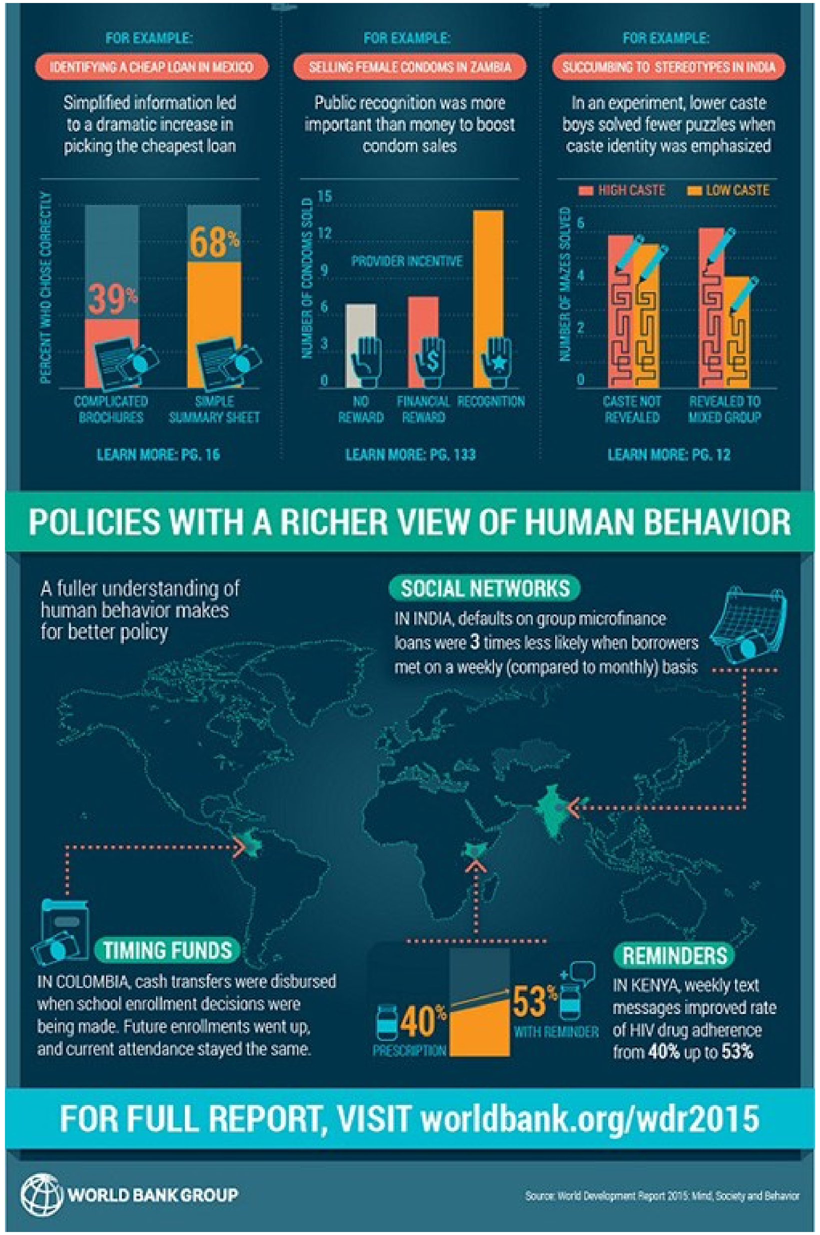
SLD Lab

PROGRAM IN NEUROSCIENCE & COGNITIVE SCIENCE

System 1 and System 2 - A big idea

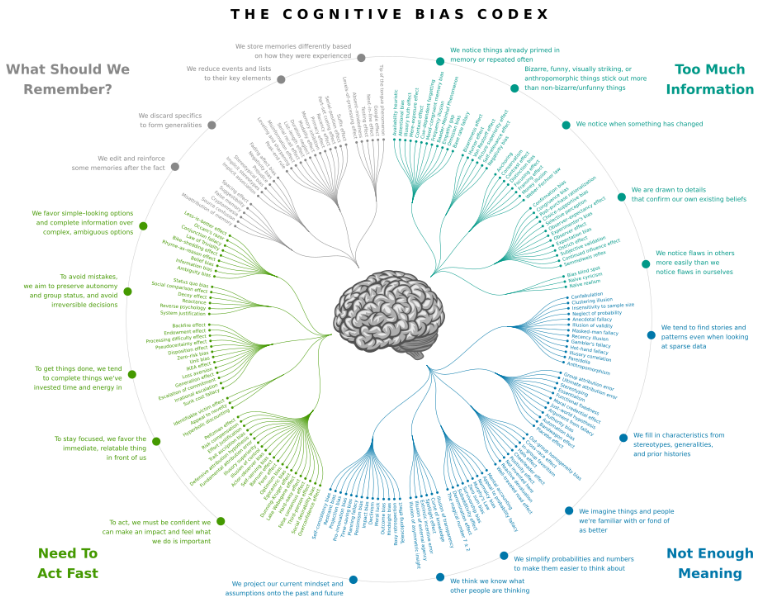
In 2015, the *World Bank* called on decision-makers to use System 2 thinking in order to avoid the errors associated with System 1 thinking



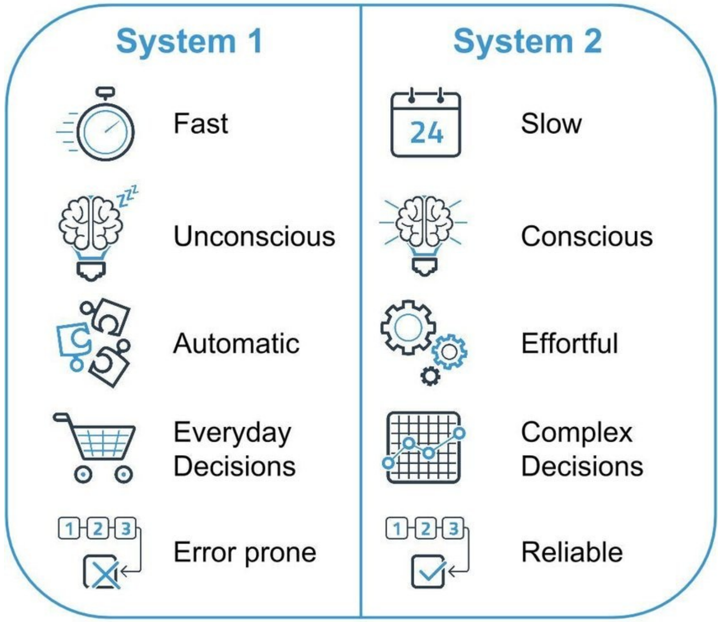


White; World Bank, *World Development Report 2015: Mind, Society and Behaviour*

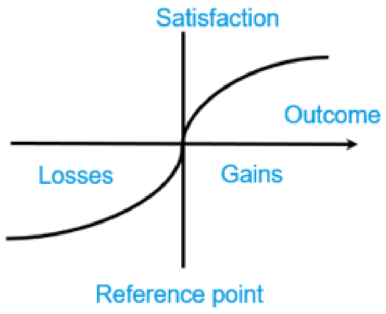
Ecological rationality
- Kahneman & Tversky

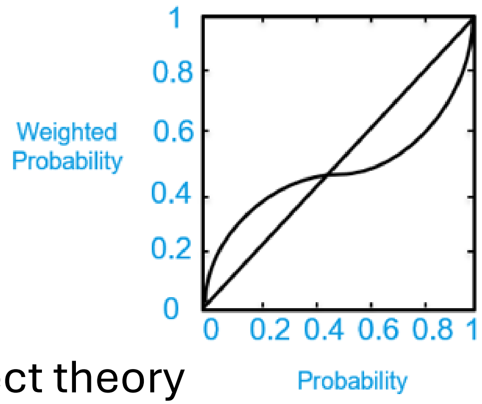


Heuristics and biais
1970's



Dual-process theory
1990-2000





Prospect theory
1979

System 1 and System 2 - Framework

Trait	Type 1 (Fast)	Type 2 (Slow)
Consciousness	Unconscious	Conscious
Intentionality	Unintentional	Intentional
Efficiency	Efficient (low cognitive cost)	Inefficient (high cognitive cost)
Controllability	Uncontrolable	Controlable

Postulates:

- Mental processes naturally fall into **two distinct types**
- **Knowing one trait** (e.g. a process is unconscious) **helps deducing the other traits** (e.g. therefore is also unintentional, efficient and uncontrollable).
- This grouping **reflects the human cognitive architecture**

Melnikoff et Bargh (2018):

- Many psychological processes **combine traits from both types**. E.g., Intentional but unconscious: driving, typing, playing the piano
- The traits themselves are not **unitary**, each breaking down into **sub-components** that don't always coincide. E.g., controllability is heterogeneous (modulable vs preventable)

Dual-process theories

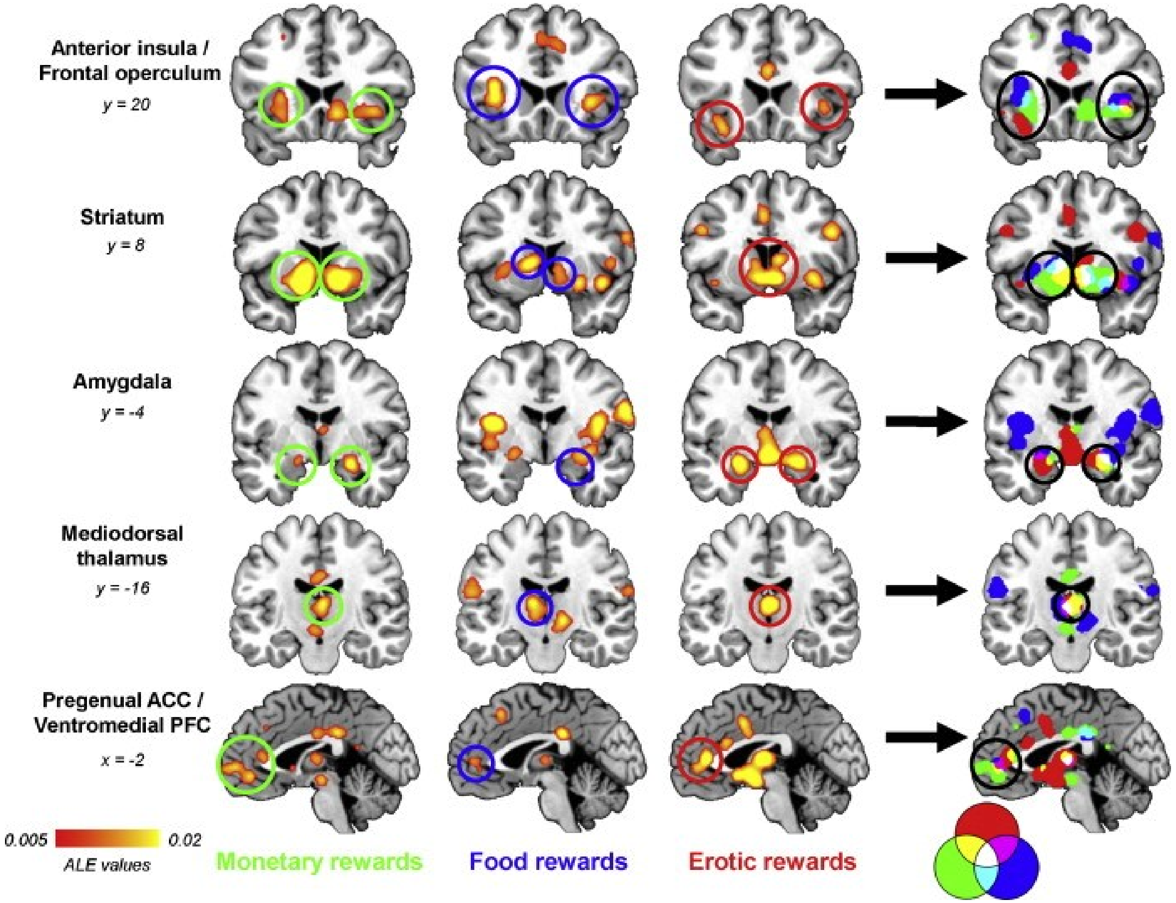
Dual-process theories of reasoning originates from classical views of rationality & philosophy: ***passion vs reason***, vs *Damasio*

Some other modern dual-process theories:

- Emotions: hot vs cold
- Action-selection: habitual vs goal-directed
- Persuasion: central route vs peripheral route
- Reinforcement learning: model-free vs model-based
- Decision: visceral/somatic markers vs high-level cognition

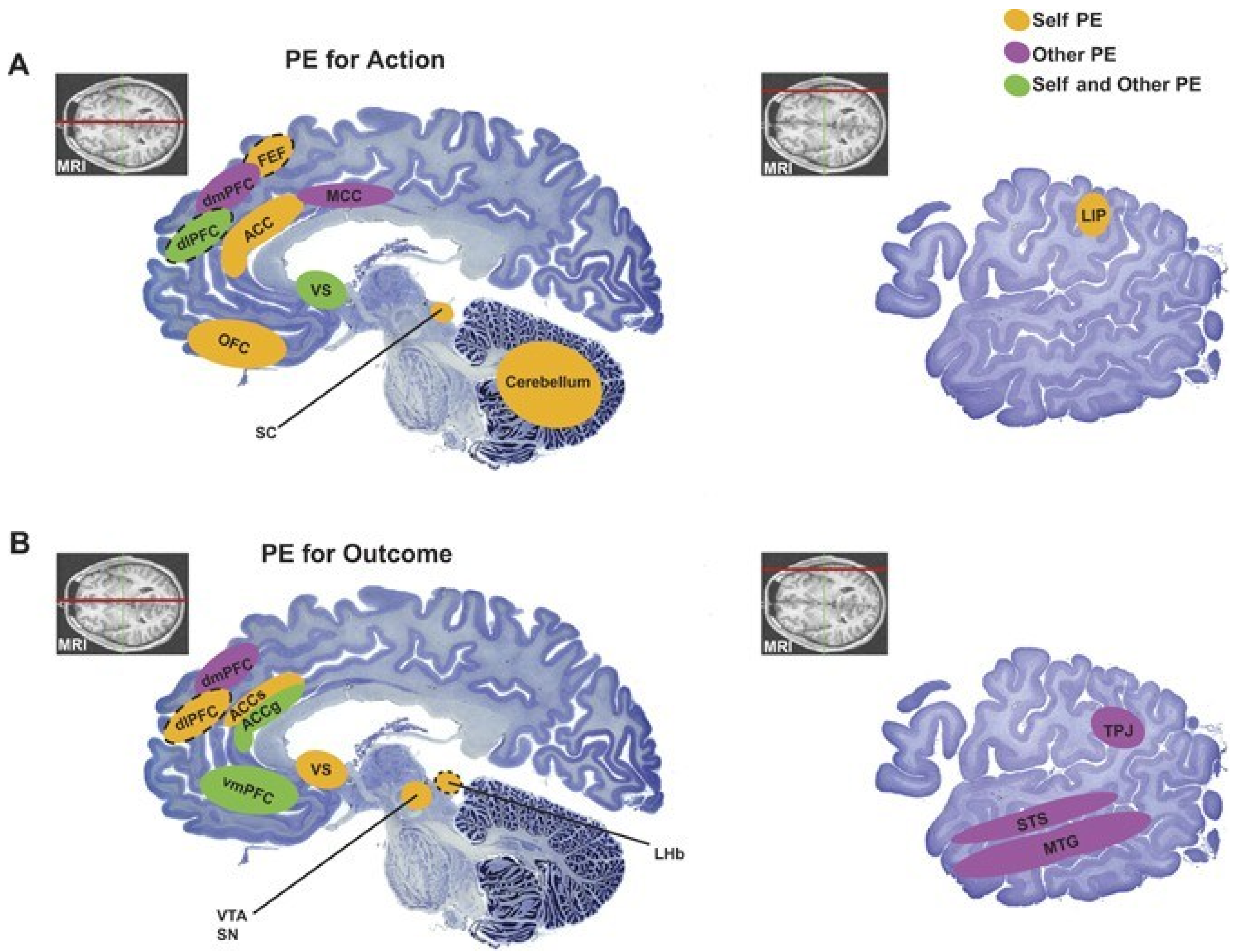
Problem: easy to “maintain” beliefs about theories when they not properly constrained:
(e.g., modularity: Fodor’s 9 rules -> 9-n rules depending on authors; system 1/system 2 traits revised)

Emotion vs Reason?



Common reward circuit during outcome

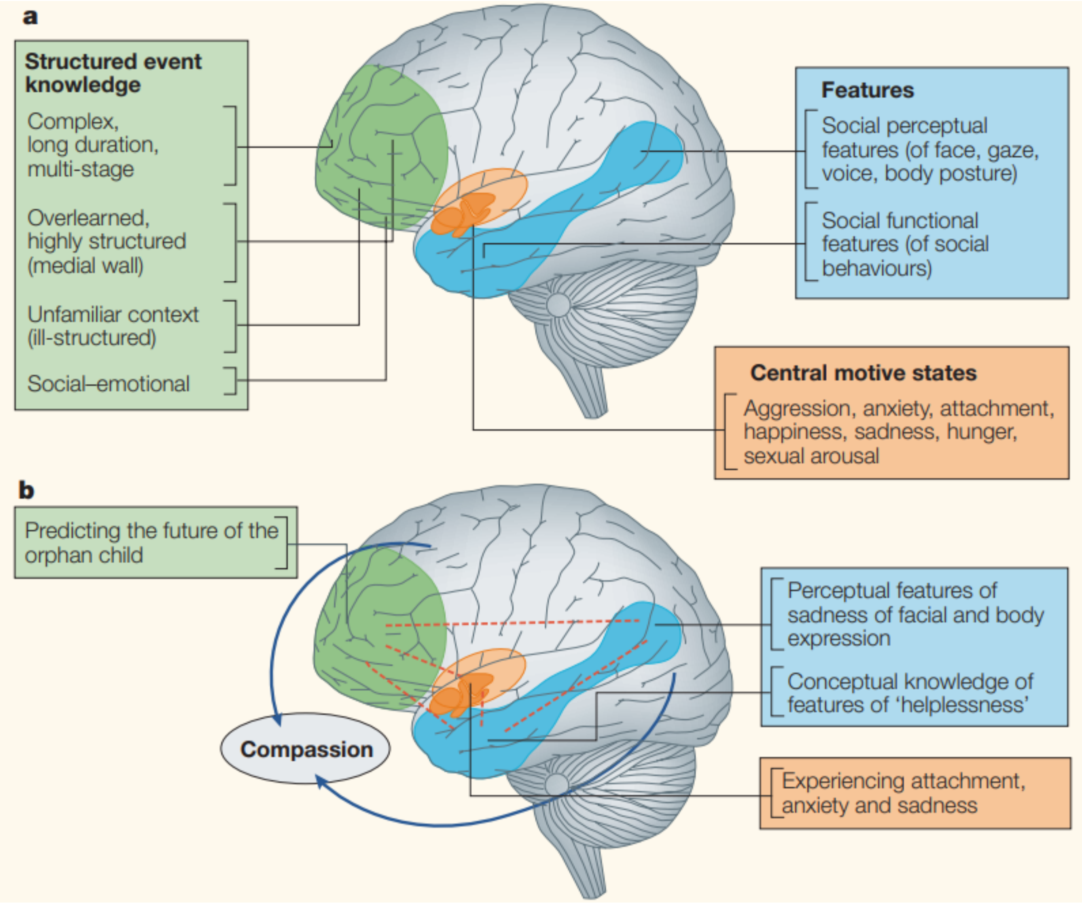
Emotion vs Reason?



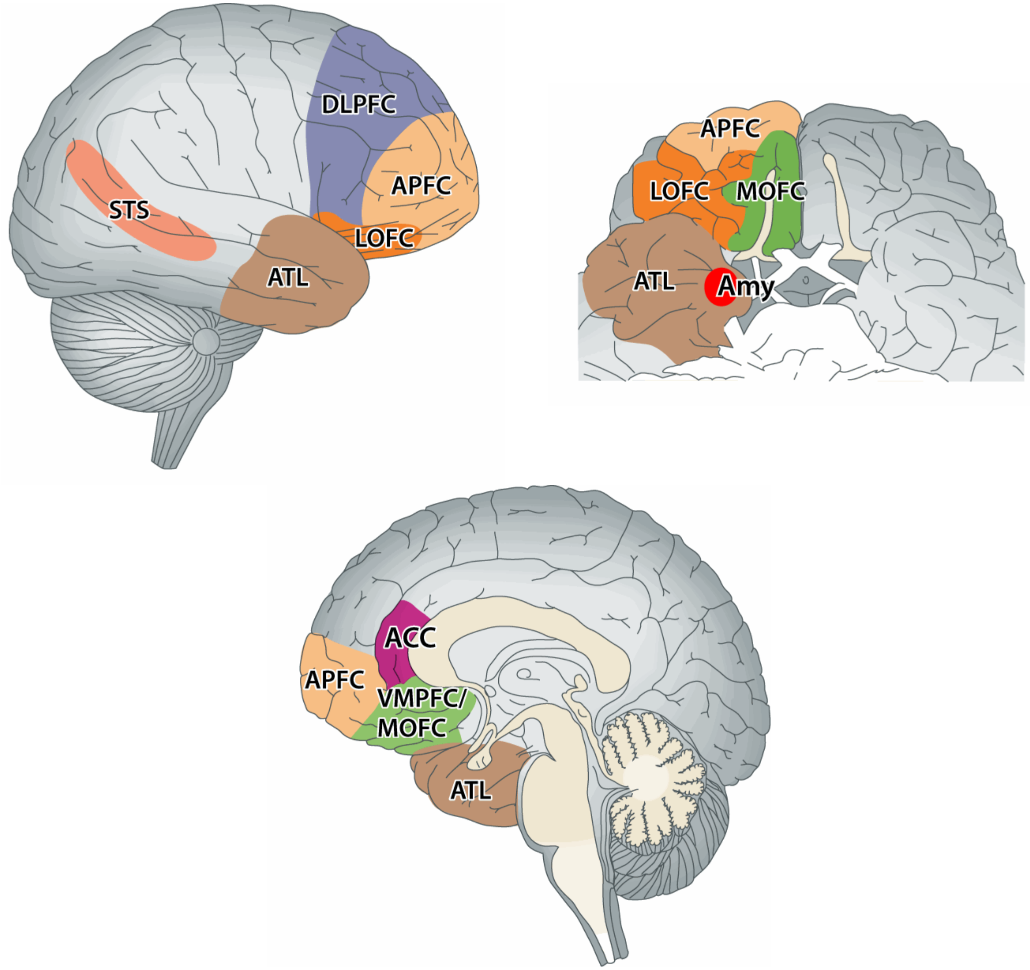
Social prediction

Emotion vs Reason?

Moll et al., 2005. *Nature reviews neuroscience*
Forbes et Grafman, 2010. *Annual review of neuroscience*

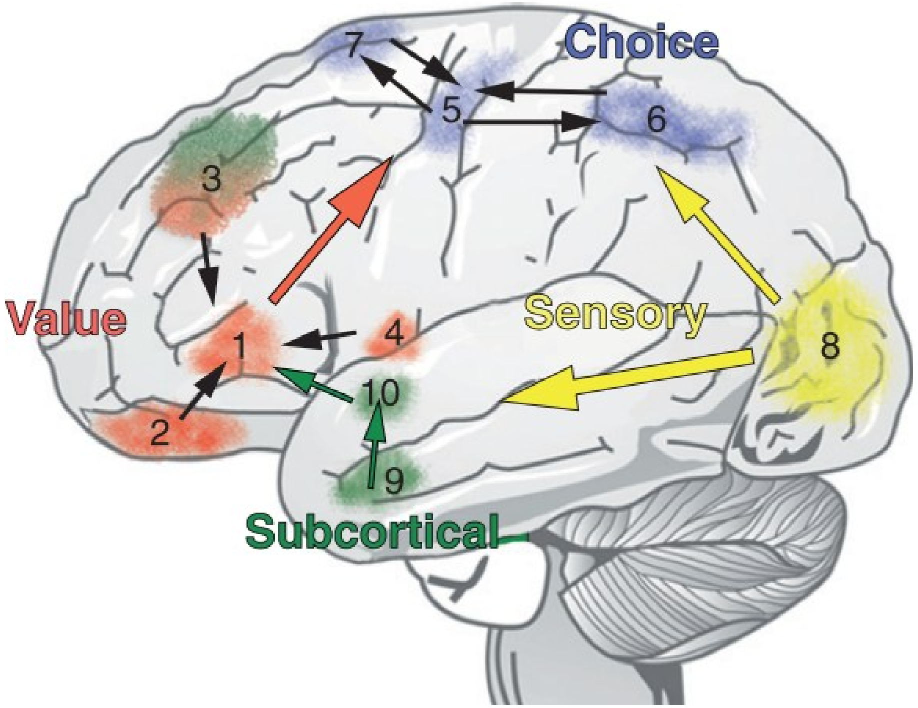


Moral sentiments/processing



Moral judgments

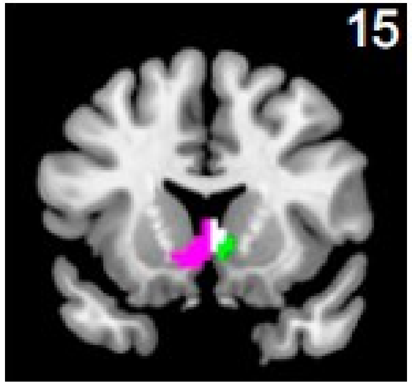
Emotion vs Reason?



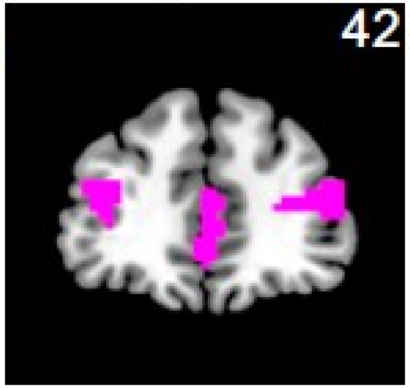
Levy & Glimcher, 2012. *Current Opinion in Neurobiology*

Valuation network

Subjective value of information



Striatum



VMPFC & Lat. PFC

Kobayashi & Hsu, 2019. *PNAS*

Classical rationality

Homo economicus

- Rational agents
- Perfectly informed
- Maximize their utility

Rational choice theory

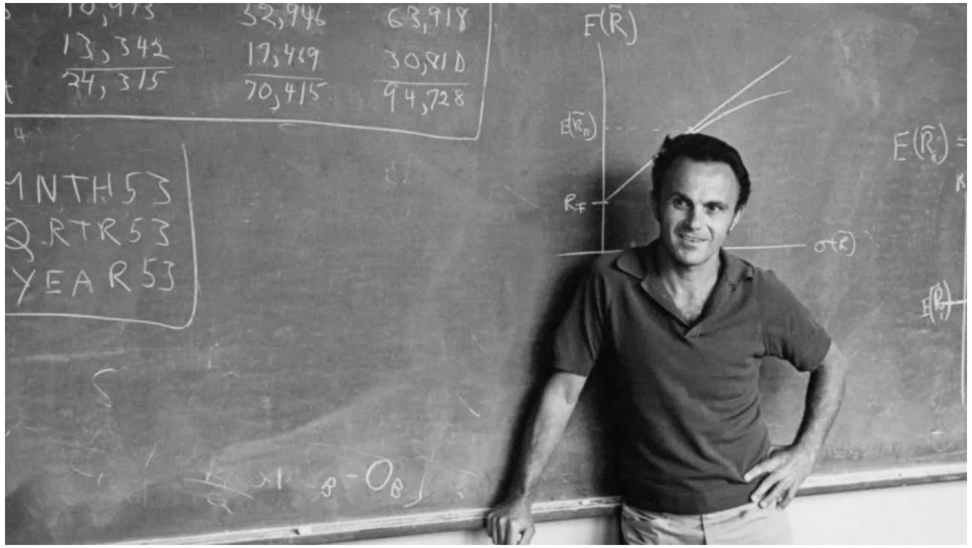
- Coherent (ordered) preferences
- Rational thinking leads to choices aligned with preferences
- Group behavior reflect the aggregate of individual behaviors (efficient allocation)

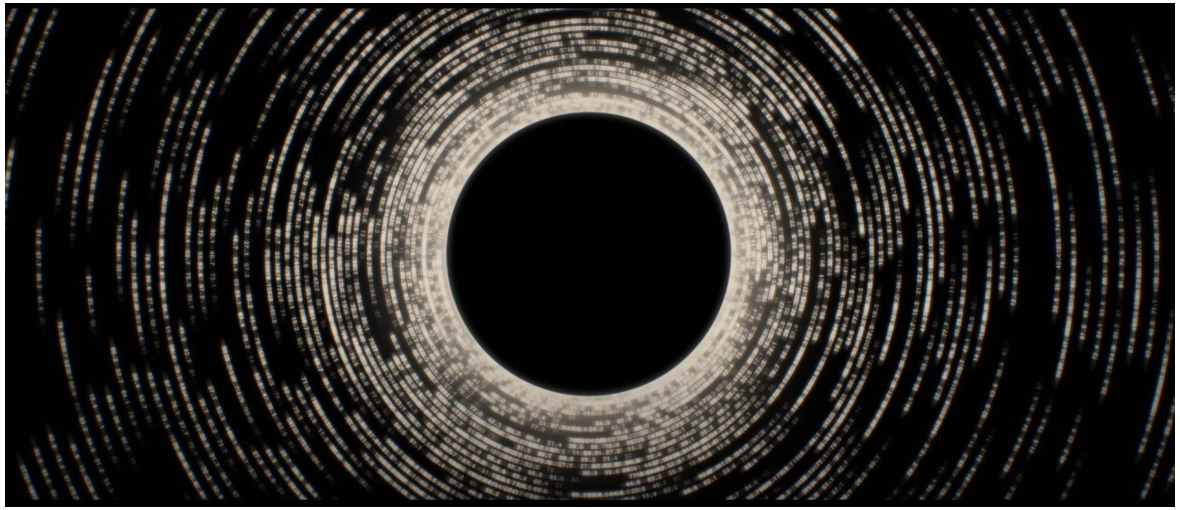
Mill, Pareto, Samuelson, etc.

Classical rationality scaled to markets

Efficient market:

- Many agents interact within markets
- In a given market, all information is directly available (efficient market, no information asymmetry)
- Markets foster rationality -> choices reveal preferences
- Free and perfect competition -> optimal equilibrium (efficient allocation), triumph of the most efficient





Marketplace of ideas:

Analogy of the free market applied to ideas and expression. When information is abundant and competition between ideas is free:

- Higher quality information gets the upper hand
- Truth prevails
- Quality of judgments improve

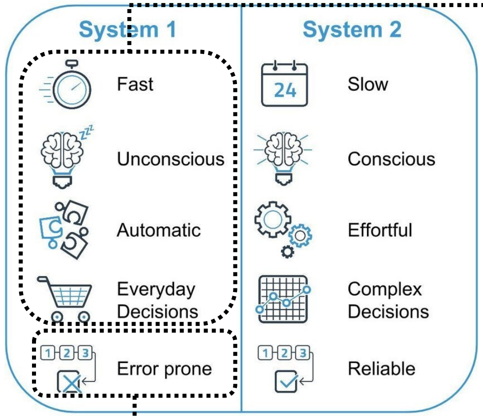
Hayek, Mill, Pareto, Fama vs Schiller

Ecological rationality / Bounded rationality

3 modern rationality traditions:

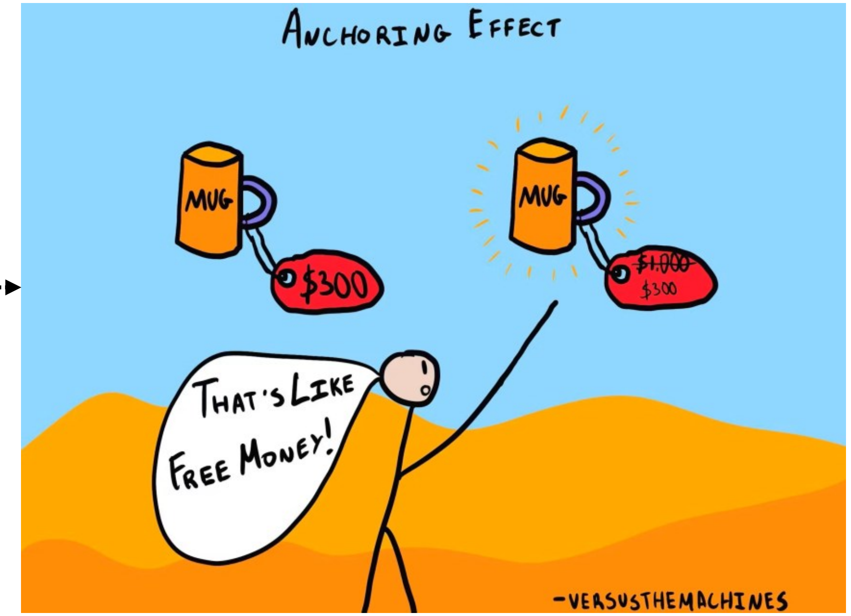
- Logical rationality (post-WWII, tied to game theory): **formal rules** like expected utility maximization, Bayesian updating, Nash equilibria
- Heuristics-and-biases program (70s): humans **follow rules of logical rationality**, but they **tend to deviate** from them; these deviations are called biases; biases justify interventions (e.g., nudges)
- Ecological rationality (90-2000): **rationality is not consistent with classical axioms** (transitivity, coherence, stability) but rather **bounded** by biological and ecological constraints: uncertainty, task complexity and available cognitive resources

From system 1
to heuristics and biases



Heuristics

- **Simple, fast, frugal** cognitive strategies
- Often ignore part of the information to find a good-enough (rather than perfect) solution



Biases

- Often seen as **error or systematic deviations** between a human judgment and a norm of rationality (e.g., laws of probabilities or logic)

Why employ heuristics: The cost-accuracy trade-off

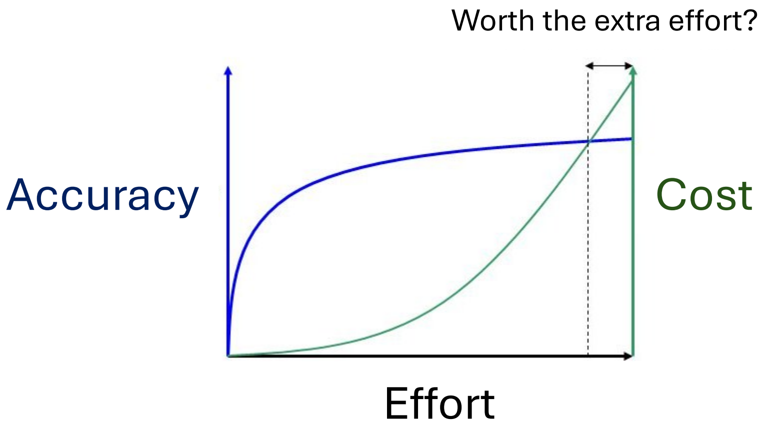
"Heuristics and Bias": the idea of fast, frugal but low-quality mental software.

1) **Heuristics are always the 2nd best choice**
2) **We use them because of cognitive limitations**
3) **More informations, more computation and more time would always be preferable (1st choice)**

- Based on the **cost-accuracy trade-off** hypothesis: accuracy is linked to the effort expended (information, calculations, time)
- Heuristics would save resources at the cost of quality

Principle of total evidence

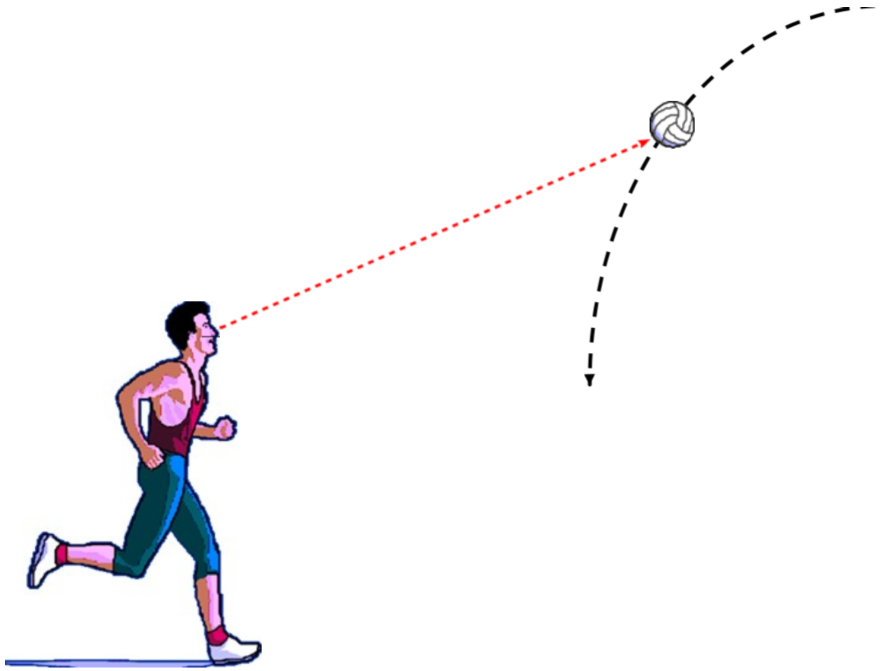
Cost-accuracy trade-off



Cost-benefit ratio to be optimized

Gigerenzer & Brighton, 2009

Are heuristics the 2nd best choice?



As efficient and less demanding than a calculation based on more information (e.g. differential equations)

Heuristics are functional responses to environmental uncertainty:

- Ignore information
- Are computationally efficient (neither maximization nor optimization)
- Good enough" solutions
- Are adapted (a given heuristic is optimal in certain environmental contexts, to the detriment of others)

Gigerenzer & Brighton, 2009

Less is more
- is more always preferable?

Reasons for the cognitive system's use of heuristics:

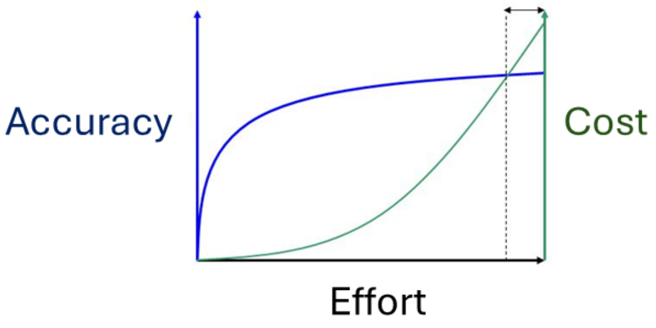
1. Cost-accuracy trade-off (cost savings)

2. « Less is more » (selective ignorance)

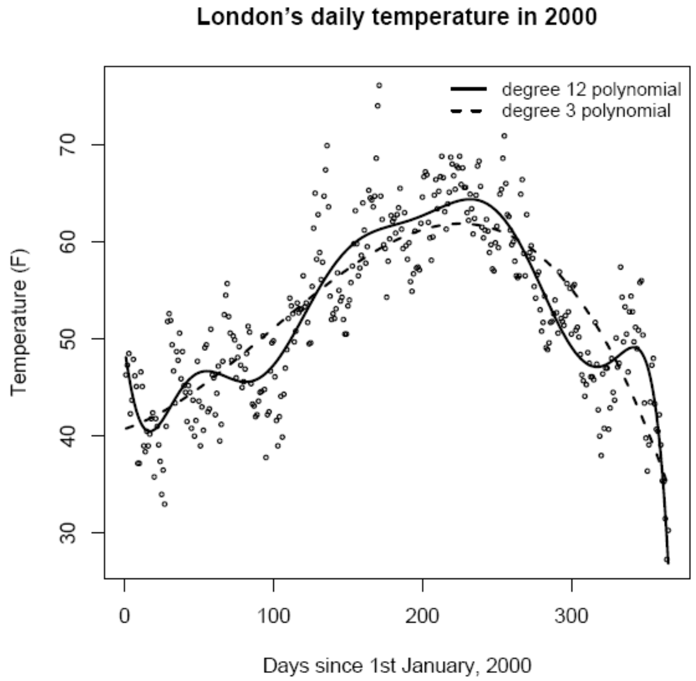
There is a point at which **more** information (indices, weights or dependencies between indices) or more calculations can become **detrimental, regardless of cost.**

More can reduce precision

Cost-accuracy trade-off



Less is More

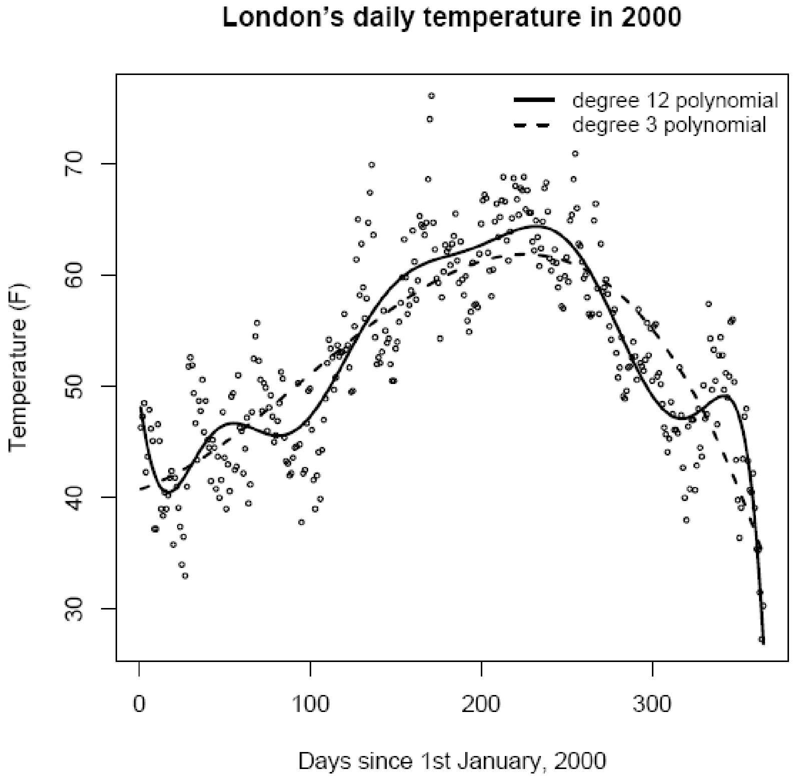


Gigerenzer & Brighton, 2009

Less is more
- cognitive limitations?

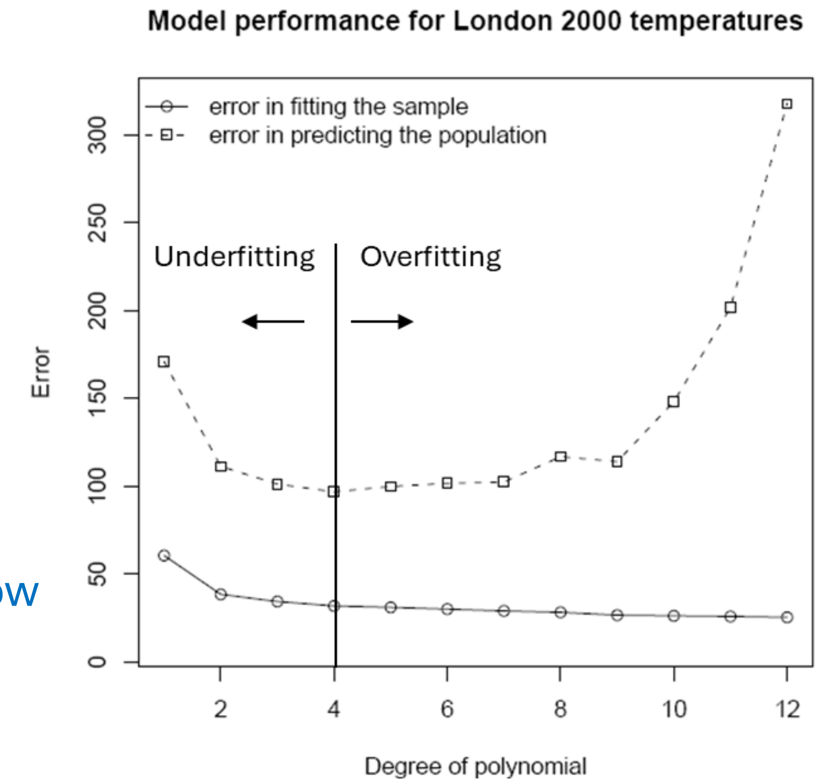
A model that takes all information into account (**good *fit***) does not guarantee good performance.
The model could simply absorb **unsystematic variations**.

The ability to predict unobserved events (**good prediction**) is a better indicator.
Models are predictive **because** they primarily capture systematic regularities.



Polynomial 3 (low variance):
Low effort, medium accuracy, strong prediction

Polynomial 12 (high variance):
High effort, medium accuracy, low prediction



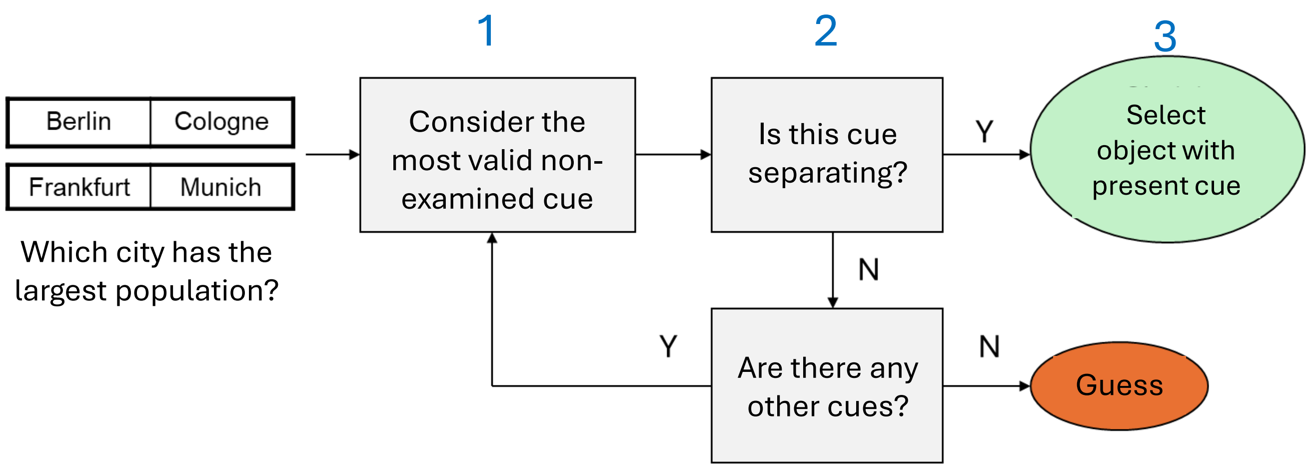
Gigerenzer & Brighton, 2009

Less is more
- example: *Take the best*

One-good-reason family heuristic:
Uses binary indices (1 vs 0) ordered by predictive validity.

Three operational rules:
1. Search rule
2. Stopping rule
3. Decision rule

City	Population	Soccer team?	State capital?	Former GDR?	Industrial belt?	License letter?	Intercity train-line?	Expo site?	National capital?	University?
Berlin	3,433,695	0	1	0	0	1	1	1	1	1
Hamburg	1,652,363	1	1	0	0	0	1	1	0	1
Munich	1,229,026	1	1	0	0	1	1	1	0	1
Cologne	953,551	1	0	0	0	1	1	1	0	1
Frankfurt	644,865	1	0	0	0	1	1	1	0	1
...	...									
Erlangen	102,440	0	0	0	0	0	1	0	0	1
Cue validities:		**0.87**	**0.77**	**0.51**	**0.56**	**0.75**	**0.78**	**0.91**	**1.00**	**0.71**



Less is more
- exemple: *Take the best*

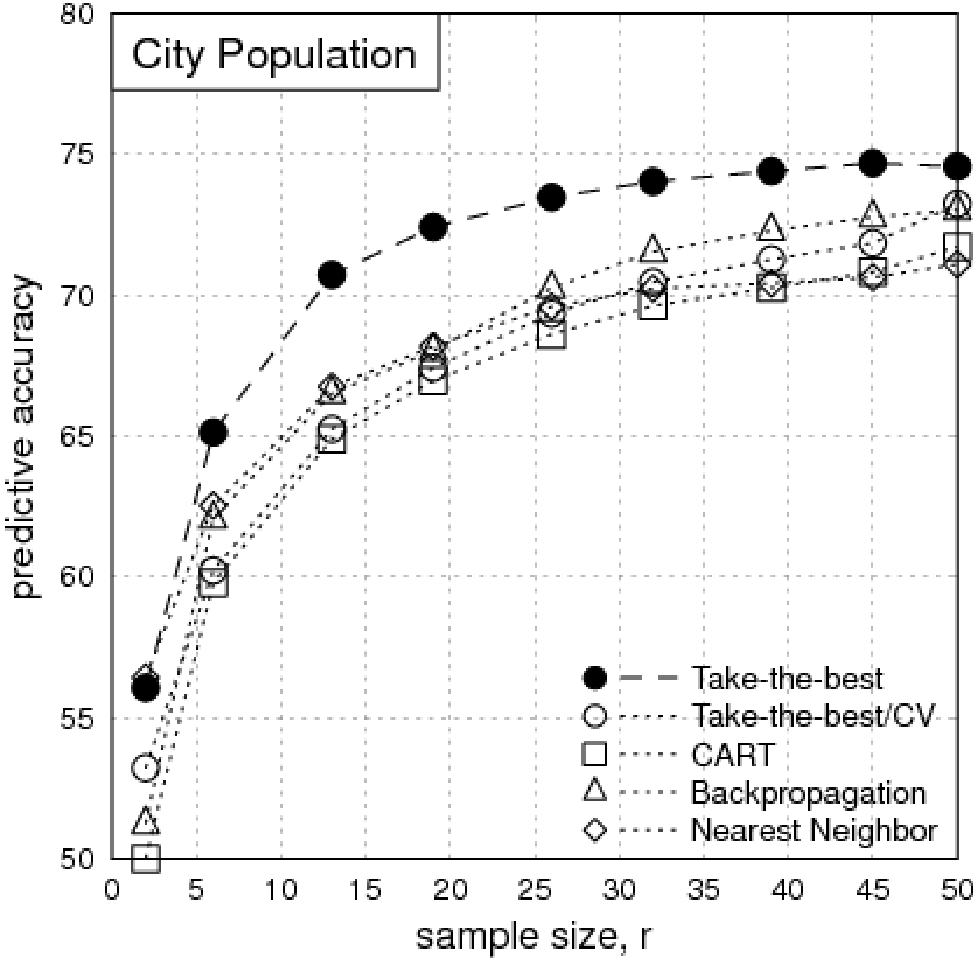
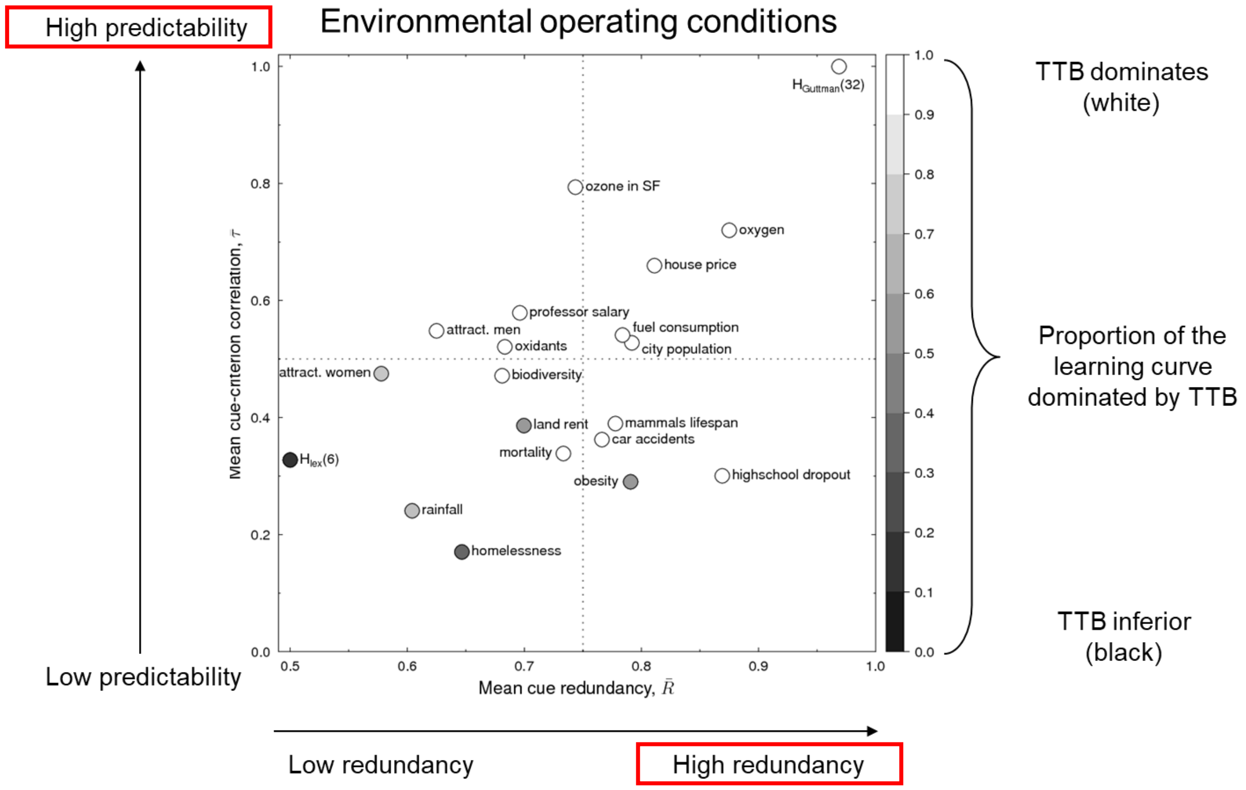


Table 5-4: Performance Across 20 Data Sets

Strategy	Frugality	Accuracy (% Correct) Fitting	Generalization
Minimalist	2.2	69	65
Take The Best	2.4	75	71
Dawes's rule	7.7	73	69
Multiple regression	7.7	77	68

Performance in 20 environments



Czerlinski, Gigerenzer, & Goldstein, 1999; Gigerenzer & Brighton, 2009

When biases lead to better inferences
(vs. complex models)

When information is scarce, degraded, uncertain, complex, noisy
When the environment is sufficiently predictable

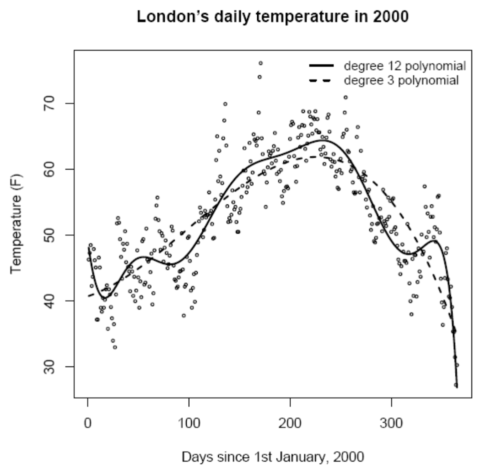
- **Predictive superiority**
- **Robustness to uncertainty**
  Ignoring information can make predictions less sensitive to noise and small samples
- **Cognitive efficiency**
  Reduces cost while maintaining sufficient performance (Martignon et al., 2008)

- **By simplifying**, heuristics introduce a bias:

→ This bias reduces the instability of predictions (variance)
→ Improves robustness and generalizability to similar situations, especially under uncertainty

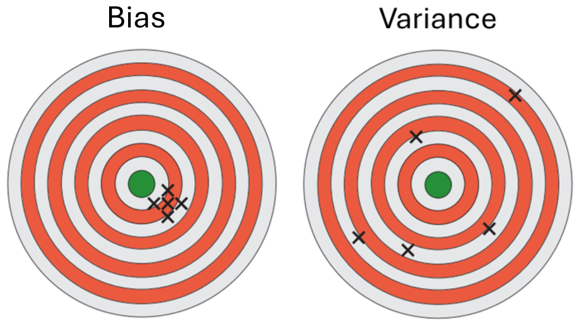
- **By complexifying**, models reduce bias but become more sensitive to noise :

→ This increases the variance of predictions
→ Reduces ability to generalize to new situations



Gigerenzer & Brighton, 2009

Prediction error in predictive models:



+ noise (ε)

Good vs Bad

Griffits et Tenenbaum, 2006. *Psychological Science*
Callaway et al., 2018. *Proceedings of the Annual Meeting of the Cognitive Science Society*
Melnikoff et Bargh, 2018. *Trends in Cognitive Sciences*

Some dual-process theories don't make sense anymore (reason vs emotions), some do.

Judgments typically associated with System 1 tend to be optimal:

- Everyday decisions approach the performance of an ideal Bayesian observer (Griffiths & Tenenbaum, 2006)
- Regular planning of tasks achieve ~86% of the optimal trade-off between decision quality and cognitive cost (Callaway et al., 2020)

Judgments typically associated with System 2 can lead to motivated beliefs:

Calling Type 2 thinking good is to champion motivated reasoning, the domain of self-serving rationalizations and of finding creative self-serving justifications [...]. (Melnikoff et Bargh, 2018)